

AI Agent Frameworks

You've heard these names everywhere

LangChain. AutoGen. CrewAI.

But what exactly are they?





They are Python libraries for building AI agents

Pre-built components, agent classes, and tool integrations — so you don't build everything from scratch.

Pre-built Components

Ready-made building blocks for common agent tasks

Agent Classes

Structured abstractions for defining agent behaviour

Tool Integrations

Connect to APIs, databases, and external services instantly



MOST POPULAR

LangChain

The most widely used framework for building AI agents. Create agents, chains, and tool integrations with minimal code.

Chains

Do multiple things in a specific order, one after another.

Agents

Autonomous decision-making loops with tool access built in

Integrations

Hundreds of ready-made connectors for tools and data sources

AutoGen – multi-agent collaboration

Define multiple agents with roles and let them collaborate automatically. AutoGen orchestrates the conversation between agents so complex tasks get solved through teamwork.

1

Define Agents

Assign each agent a name, role, and system prompt

2

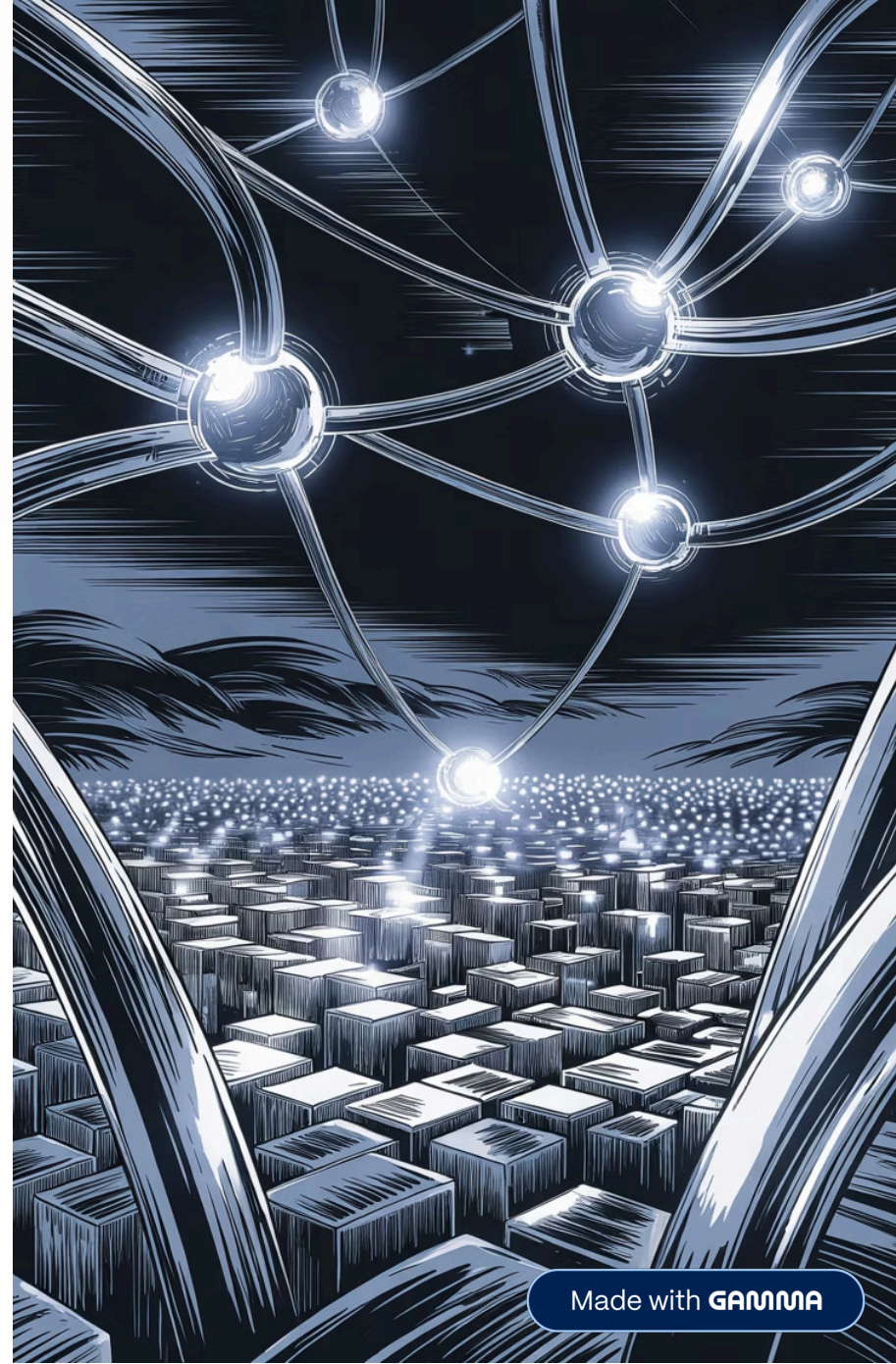
Set the Task

Give the group a goal or problem to solve together

3

Collaborate

Agents message each other automatically until the task is complete





ROLE-BASED

CrewAI – teams of agents

Assign roles, goals, and tasks to each agent — CrewAI handles the teamwork. Think of it as building a crew where every member knows their job.



Roles

Each agent gets a specific identity and area of expertise



Goals

Agents are driven by individual objectives that serve the crew

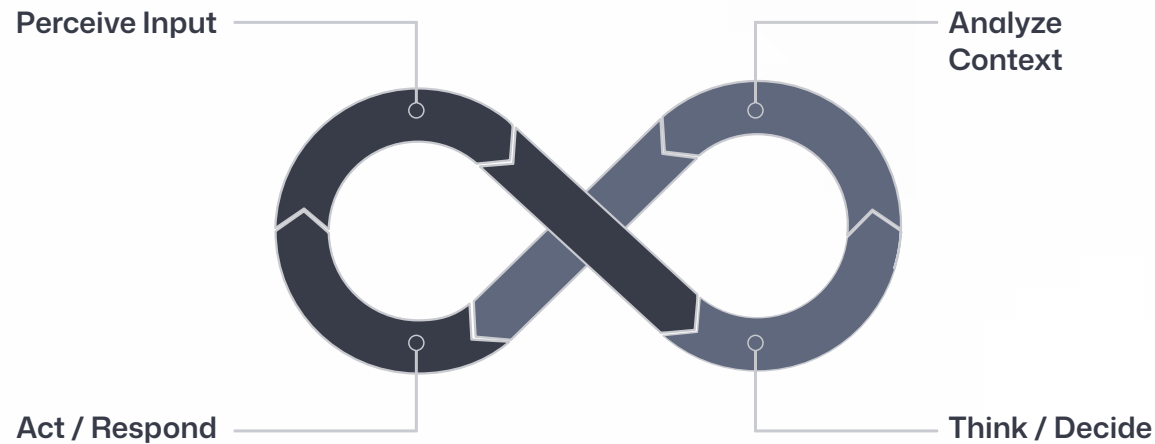


Tasks

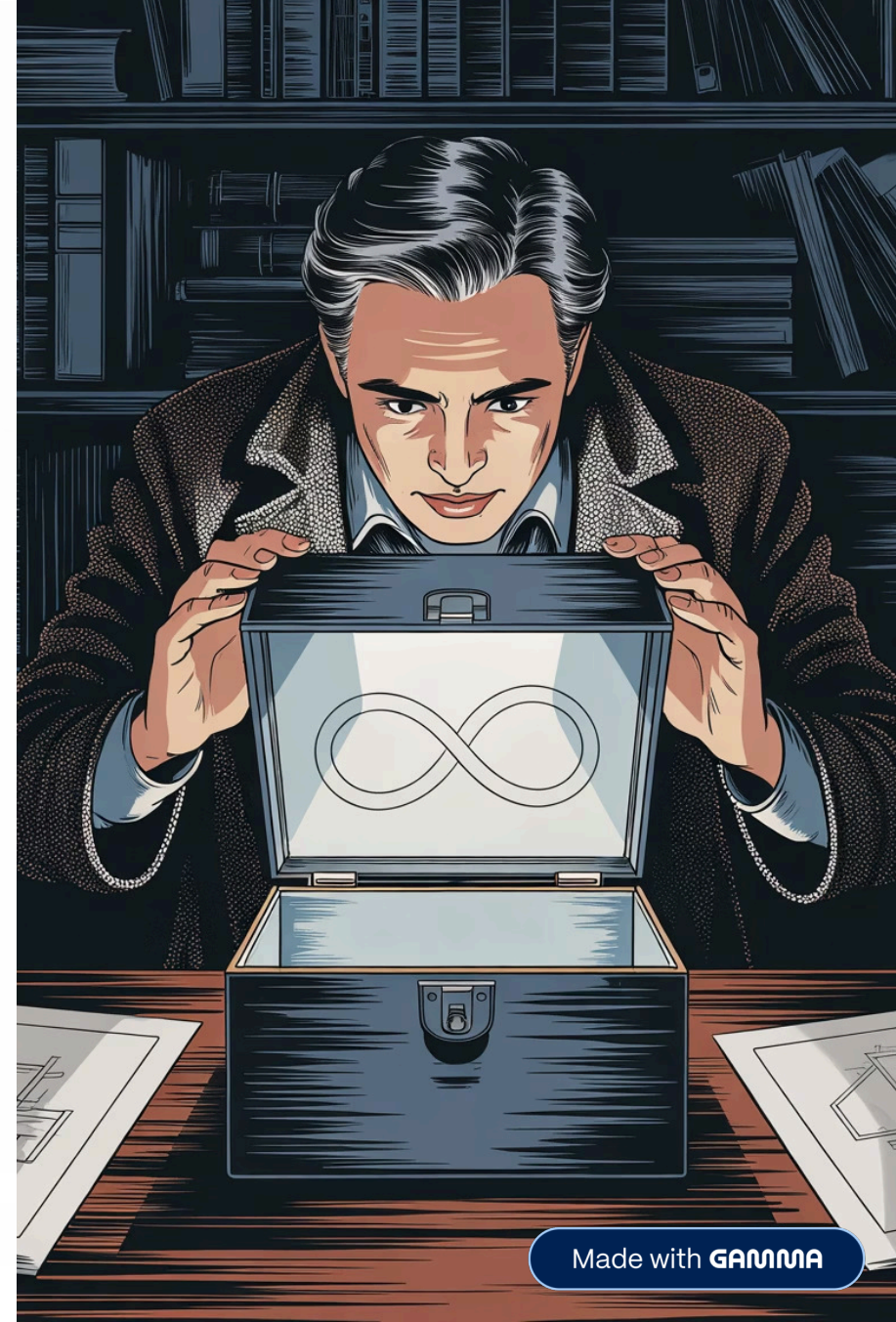
Discrete units of work assigned and tracked automatically

What's happening inside all of them?

Under the hood — every framework is built on the same agent loop you already learned. The abstractions differ, but the core engine is identical.



No matter how sophisticated the framework looks on the surface, it is always running this same fundamental cycle beneath the abstractions.





Learn manual before automatic

Build raw first — understand every line. Then any framework takes one day to learn. Jumping straight to a framework means you are always guessing what it does internally.

1

Build Raw

Write the agent loop yourself — every line, every decision

2

Understand Deeply

Know exactly what happens at each step of the loop

3

Pick Any Framework

With deep knowledge, any framework becomes obvious in a day



At the end of this course

I will show you LangChain doing exactly what we built — and you will understand it **immediately**. The code will look familiar because the logic is the same. Only the syntax changes.